CISS245: Advanced Programming
Quiz q04

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Q1. Either write down the output of this code fragment or write ERROR if it is not valid C++.

```
int x = 0, y = 42, z = 99;
if (x < 0)
{
    int y = 1, z = 100;
    if (x == 0)
    {
        int y = 2;
        if (y < 1)
        {
            std::cout << z;
        }
        else
        {
            std::cout << z;
        }
    }
    else
    {
        int y = 3;
        if (y < 2)
        {
            std::cout << z;
        }
        else
        {
            std::cout << z;
        }
    }
}
else
{
    int y = 4, z = 200;
    if (x == 0)
    {
        if (y < 3)
        {
            std::cout << z;
        }
        else
        {
            std::cout << z;
        }
    }
    else
    {
        if (y < 4)
        {
            std::cout << z;
        }
```

```
        else
        {
            std::cout << z;
        }
    }
}
```

Answer:

```
200
```

Q2. Rewrite the following function so that the if-else statement is replaced by the ternary operator. The function should contain only one statement.

```
int sign(double x)
{
    int ret;
    if (x >= 0)
    {
        ret = 1;
    }
    else
    {
        ret -1;
    }
    return ret;
}
```

Answer:

```
int sign(double x)
{
    return (x >= 0 ? 1 : -1);
}
```

Q3. Write a function

```
int numdigits(int n);
```

that returns the number of digits in `n`. For instance if `n` is 0 or 5 or 9, the function returns `1`. If `n` is 23 or 42 or 99, the function returns `2`. If `n` is -123 or -243 or -798, the function returns `3`. Etc.

Answer:

```
int numdigits(int n)
{
  if(n == 0)
  {
    return 1;
  }
```

```
  int count = 0;

  n = n < 0 ? n *= -1 : n;

  while(n > 0)
  {
    n /= 10;
    count++;
  }

  return count;
}
```

Q4. Write a function

```
bool isprime(int n);
```

that returns `true` if and only if `n` is a prime. For instance if `n` is 2, 3, 5, 7, 11, 13, 17, 19, 23, or 29, the function returns `true`. If `n` is 0, 1, 4, 6, 8, 9, 10, 12, 14, or 15, the function returns `false`.

Answer:

```
bool isprime(int n)
{
  if(n <= 1)
  {
    return false;
  }

  for(int i = 2; i * i <= n; i++)
  {
    if(n % i == 0)
    {
      return false;
    }
  }

  return true;
}
```

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.